

Complete Project Setup, Local Reconstruction & Deployment Guide

A Complete, Standalone Manual to Set Up, Run, Test, and Deploy NexusResearch on Any Windows Computer from Scratch Without AI Assistance

[AUTHENTIC SOURCE OF TRUTH] This document is authored directly from the NexusResearch production repository. Every command, port, environment variable, database table, API endpoint, and deployment step in this guide has been verified against active code. No real API keys, passwords, or OAuth secrets are published in this guide; placeholder formats are provided throughout.

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Target Environment: Windows 10 / Windows 11 (64-bit)

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Repository Source: <https://github.com/TejeshwarDivekar/NexusAI.git>

1. Project Overview & Capabilities

NexusResearch is a full-stack, enterprise-grade AI research assistant engineered to automate grounded scholarly research, evidence synthesis, conflict auditing, and publication-ready academic document compilation. Unlike generic LLM wrappers that hallucinate facts or generate fabricated statistics, NexusResearch enforces a strict 7-stage deterministic pipeline that connects directly to real peer-reviewed registries and authoritative web databases.

Key Problems Solved:

- **Academic Hallucinations:** Every factual claim synthesized by the system is linked to a verified sentence-level quote with an exact bracketed citation ([1], [2]).
- **Research Fragmentation:** Aggregates findings concurrently across OpenAlex, PubMed, Europe PMC, arXiv, Crossref, and Wikipedia into a single unified workspace.
- **Manual Paper Formatting:** Automatically exports validated research findings into double-column IEEE standard Word documents (.docx) and formal academic PDFs (.pdf).
- **Data Governance & Multi-Tenancy:** Ensures strict database-level isolation of user projects, research histories, and uploaded context documents.

2. System Requirements & Prerequisites Table

Software	Required Version	Purpose in NexusResearch	Verification Command
Windows OS	10 / 11 (64-bit)	Host operating system	systeminfo
VS Code	Latest Stable	Development IDE & terminal	code --version
Git for Windows	v2.40.0+	Version control & cloning	git --version
Node.js	v18.17+ / v20+	Next.js frontend runtime	node --version
npm	v9.0+ / v10+	Frontend package manager	npm --version
Python	v3.10.x - v3.12.x	FastAPI backend & doc gen	python --version
pip	Latest	Python package manager	pip --version
Google Gemini API	API Access Key	Synthesizing research answers	N/A (Web Portal)

3. Tool Installation (VS Code, Git, Node.js, Python)

1. **VS Code:** Download from <https://code.visualstudio.com>. During setup, enable 'Add to PATH'. Open terminal via Ctrl + `.
2. **Git for Windows:** Download from <https://git-scm.com/download/win>. Verify with `git --version`.
3. **Node.js & npm:** Download v20 LTS from <https://nodejs.org>. Verify with `node --version` and `npm --version`.
4. **Python 3.10+:** Download from <https://python.org>. CRITICAL: Check 'Add python.exe to PATH'. Verify with `python --version`.

4. Obtaining the Code & Repository Structure

```
# Clone repository from GitHub
git clone https://github.com/TejeshwarDivekar/NexusAI.git
cd NexusAI
code .
```

5. Installing Dependencies & Configuring Environment

Install frontend packages (npm) and backend packages (pip):

```
# 1. Install frontend packages in root
npm install

# 2. Install backend Python packages
cd backend && pip install -r requirements.txt && cd ..
```

Master Environment Variables Table:

Variable Name	Req?	Target	Purpose	Development Value
AUTH_SECRET	Yes	.env.local	NextAuth session signing secret	Run: npx auth secret
AUTH_TRUST_HOST	Yes	.env.local	Trust proxy host headers	true
NEXTAUTH_URL	Yes	.env.local	Public base URL	http://localhost:3000
BACKEND_INTERNAL_URL	Yes	.env.local	Proxy target for FastAPI	http://127.0.0.1:8000
GOOGLE_GENERATIVE_AI_API_KEY	Yes	Both	Gemini LLM API Key	From aistudio.google.com
SECRET_KEY	Yes	backend/.env	FastAPI JWT signing key	32+ character string
DATABASE_URL	Yes	backend/.env	SQLite connection string	sqlite:///./nexusai_research.db

6. Running Backend & Frontend (Dual-Terminal Workflow)

NexusResearch requires two active terminal sessions in VS Code:

```
# TERMINAL 1 — Start FastAPI Backend Server (Port 8000)
python backend/run.py
# Expected: Uvicorn running on http://0.0.0.0:8000

# TERMINAL 2 — Start Next.js Frontend Server (Port 3000)
npm run dev
# Expected: Ready on http://localhost:3000
```

7. Verifying Features & Testing the Application

- 1. **Authentication:** Open `http://localhost:3000` and log in via Google OAuth or local credentials.
- 2. **Research Inquiry:** Enter 'What is artificial intelligence?' and observe Stage 1 to 7 progress stream.
- 3. **Evidence Matrix Interaction:** Click an evidence item in the right panel. Inspect the center detail explanation, then click '<— Back to Answer' to verify instant scroll restoration.
- 4. **Document Export:** Click 'Export IEEE (.docx)' and 'Export PDF (.pdf)' to download publication-grade reports.
- 5. **Automated Test Suite:** Run `python -m pytest backend/tests -v` (30/30 tests pass).
- 6. **Production Build Check:** Run `npm run build` (Next.js 16.3.1 compiles with 0 errors).

8. Printable Fresh-Laptop Setup Checklist

- [] 1. Windows OS updated and VS Code installed with PowerShell terminal
- [] 2. Git for Windows installed (`git --version` verified)

- [] 3. Node.js v20.x and npm installed (`node --version` verified)
- [] 4. Python 3.10+ installed with 'Add to PATH' (`python --version` verified)
- [] 5. Repository cloned via `git clone https://github.com/TejeshwarDivekar/NexusAI.git`
- [] 6. Frontend packages installed (`npm install`)
- [] 7. Backend packages installed (`cd backend && pip install -r requirements.txt`)
- [] 8. Root `.env.local` created with valid `AUTH\_SECRET` and `GOOGLE\_GENERATIVE\_AI\_API\_KEY`
- [] 9. Backend `backend/.env` created with `SECRET\_KEY` and `GOOGLE\_API\_KEY`
- [] 10. Backend started (`python backend/run.py`) on port 8000
- [] 11. Frontend started (`npm run dev`) on port 3000
- [] 12. Health check verified at `http://127.0.0.1:8000/api/v1/health`
- [] 13. UI verified at `http://localhost:3000` with test research query
- [] 14. Evidence matrix inspection and Back to Answer verified
- [] 15. IEEE Word (.docx) and PDF (.pdf) downloads verified
- [] 16. Backend test suite passed (`python -m pytest backend/tests -v`)
- [] 17. Frontend production build verified (`npm run build`)

9. Rapid Reconstruction & Security Governance

[SECURITY MANDATE] CRITICAL SECURITY GOVERNANCE MANDATE: Never commit `.env.local`, `backend/.env`, or real API keys to GitHub. Always verify your staged changes with `git status` before committing. Keep production credentials safely managed inside cloud platform settings (e.g. Railway Variables).